Mechanics

Kinematics - Problems in kinematics

1. $t=0$ leaves nest

$$s=\frac{1}{10}(t^4-20t^3+100t^2)$$

a) At $t=10$ $s=0$ the bird has returned to the nest.

b) instantaneous when $v=0$

$$v=\frac{ds}{dt}$$

$$\frac{1}{10}(4t^3-60t^2+200t)=0$$

$$\tfrac{1}{10}t(4t^2-60t+200)=0$$

$\frac{1}{10}t=0$ $\quad 4t^2-60t+200=0$

$t=0$ $\quad t^2-15t+50=0$

$t=5 \quad t=10$

$$s=\frac{1}{10}(t^4-20t^3+100t^2)$$

$$s=\frac{1}{10}(5^4-20(5)^3+100(5^2))=62.5\text{m}$$

2. $g=9.8\,ms^{-2}$

$$s=\tfrac{1}{2}t^2(t^2-2t+1)$$

$$s=\tfrac{1}{2}t^4-t^3+\tfrac{1}{2}t^2$$

$$\frac{ds}{dt}=v=\frac{4}{2}t^3-3t^2+t$$

$$v=2t^3-3t^2+t=0$$

$$t(2t^2-3t+1)=0$$

$t=0$

$$2t^2-3t+1=0$$

$$2t^2-2t-t+1=0$$

$$2t(t-1)-1(t-1)$$

$$(2t-1)\quad(t-1)$$

$2t=1 \qquad t=1$

$t=\tfrac{1}{2}$

$t=0 \quad x=0$

$t=\tfrac{1}{2} \quad x=\frac{1}{32}$

$t=1 \quad x=0$

$t_1-t_{1/2}=\frac{1}{32}$

$t=2 \quad x=2$

$\frac{1}{32}+\frac{1}{32}+2=$

$2\frac{1}{16}=2.063$

c) $\frac{1}{2}t^2$ = always positive

4. $V=(7-2t)(t+2)$

$\int (S) \xrightarrow{\frac{dy}{dx}} (V) \xrightarrow{\frac{dy}{dx}} (a)$

$a=0$

$V=7t+14-2t^2-4t$

$V=3t-2t^2+14$

$\frac{dv}{dt}=a=-4t+3$

$-4t+3=0$

$4t=3$

$t=\frac{3}{4}$

b) $V=0$ When $t=\frac{7}{2}$

$3t-2t^2+14=0$

$t=\frac{7}{2}$ $t=-2$

Ignore negative

$S=\frac{3}{2}\times\frac{49}{4}-\frac{2}{3}\times\frac{343}{8}+14\times\frac{7}{2}$

$S=38.79$

$S=\int v\,dt$ $S=\int 3t-2t^2+14\,dt$

$\frac{3}{2}t^2-\frac{2}{3}t^3+14t$

3. a) $V=12+4t-t^2$

$V=0$

$12+4t-t^2=0$

$t^2-4t-12=0$

$(t-6)(t+2)=0$

$t=6$ $t=-2$

Ignore -ve

$a=\frac{dv}{dx}=4-2t$

$t=6$

$4-12=-8=a$

$|-8|=8ms^{-2}$

$\int v\,dt=s$

$\int t^2-4t+12\,dt$

$\left[\frac{t^3}{3}-2t^2-12t\right]_0^3$

$\left(\frac{3^3}{3}-2(3)^2-12(3)\right)-0$

$3^2-2\times9+36=9-18-36$

$=-45$

$=45$

5. $a = \begin{pmatrix} 5t \\ -15t^{1/2} \end{pmatrix}$

$t=0 \; v = 20i$

$\int 5ti - 15t^{1/2}j \, da$

$\frac{5}{2}t^2 i - \frac{30}{3}t^{3/2}j$

$v = \frac{5}{2}t^2 i - 10t^{3/2}j + Ci + Dj$

$t=0 \quad 20i = 20i + 0j$

$v = \left(\frac{5}{2}t^2 + 20\right)i - 10t^{3/2}j$

$t=4$

$v = \left(\frac{5}{2} \times 16 + 20\right)i - 10 \times 8j$

$60i - 80j$

6. $v = t^{-1/2}i - 4tj$

$t=1$ at point A

$t=4$ at point B

$\int t^{-1/2}i - 4tj \, dt$

$s = 2t^{1/2}i - 2t^2 j + Ci + Dj$

$t=1:$

$s = 2i - 2j$

$t=4:$

$s = 4i - 32j$

$\frac{dy}{dx} = \frac{-32 - -2}{4-2} = \frac{-30}{2}$

$\sqrt{2^2 + 30^2} = 2\sqrt{226} = 30.067$

7. $v = 6ti - 5t^{3/2}j$

$t=0 \quad \overrightarrow{OP} = -20i + 20j$

$\frac{dv}{dt} = 6i - \frac{15}{2}t^{1/2} = a$

$t=4$

$a = 6i - 15j$

$\int 6ti - 5t^{3/2}j \, dt$

$3t^2 i - 2t^{5/2}j - 20i + 20j$

$(3t^2 - 20)i + (20 - 2t^{5/2})j$

$(3(4)^2 - 20)i + (20 - 2(2)^5)j$

$28i - 44j$

P with initial value 5,000

$t=1$, $P=6{,}000$

Pe^{kt}

$5{,}000e^{kt}$

$5{,}000 \times e^{k} = 6{,}000$

$e^{k} = \frac{6}{5}$

$\ln e^{k} = \ln\left(\frac{6}{5}\right)$

$k = \ln\left(\frac{6}{5}\right)$

$k = 0.1823$

$5{,}000e^{5(0.1823)} = 12{,}441.6$

$5{,}000e^{t(0.1823)} = 10{,}000$

$e^{t(\ln\frac{6}{5})} = 2$

$\ln e^{t\ln\frac{6}{5}} = \ln 2$

$t(\ln\frac{6}{5}) = \ln 2$

$5{,}000e^{\ln\frac{6}{5}(t)}$

$911.607784te^{\ln\frac{6}{5}t}$

$2625.43041\,8$

$7260 = 5250 + C$

$C = 1949$

$t=2$

$P = 7{,}200$

$(2, 7{,}200)$

$\frac{7{,}200 - 5{,}000}{2-0} = \frac{2{,}200}{2} = 1{,}100$

$7{,}200 = 2m + C$

$t = \frac{\ln 2}{\ln\frac{6}{5}} = 3.8$